

**SRM Institute of Science and Technology
College of Engineering and Technology**

SCHOOL OF COMPUTING

MINI PROJECT REPORT

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Project Title : Automated Plant Water System

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Particulars	Max. Marks	Marks Obtained
		Name:
		Register No:
Review 1 and 2	05	
Demo verification & viva	03	
Project Report	02	
Total	10	

Date :

Staff Name :

Signature :

AUTOMATED PLANT WATERING SYSTEM

OBJECTIVE:

The objective of an automated plant watering system is to ensure optimal soil moisture levels by automatically providing water to plants based on their needs. This system aims to conserve water, reduce manual labor, and promote healthy plant growth. It utilizes sensors to monitor soil moisture and environmental conditions, activating the water supply only when necessary. This automation enhances efficiency, making it ideal for home gardens, farms, and greenhouses.

ABSTRACT:

This paper presents the design and implementation of an automated plant watering system aimed at optimizing plant care through precise and efficient water management. The system integrates moisture sensors, a programmable microcontroller, and a water pump to deliver tailored irrigation based on real-time soil moisture levels. By continuously monitoring the soil conditions, the system minimizes water waste and ensures that plants receive adequate hydration, promoting healthier growth. The automated solution addresses common challenges in plant maintenance, such as inconsistent watering schedules and environmental fluctuations. Initial testing demonstrates significant improvements in plant health and water conservation compared to traditional manual watering methods. This innovative approach not only enhances the convenience of plant care for users but also contributes to sustainable gardening practices. Future work will focus on expanding the system's capabilities to include remote monitoring and control via mobile applications.

INTRODUCTION:

The increasing urbanization and fast-paced lifestyles of modern society often lead to challenges in maintaining healthy indoor and outdoor plants. Traditional manual watering methods can be inconsistent and time-consuming, resulting in either over-watering or under-watering, which can adversely affect plant health. As a response to these challenges, the development of automated plant watering systems has gained traction, combining technology with horticulture to ensure optimal plant care.

Automated plant watering systems utilize sensors and microcontrollers to monitor soil moisture levels and deliver precise amounts of water when needed. This technology not only simplifies the task of plant maintenance but also promotes sustainable gardening practices by conserving water. With the ability to adjust watering schedules based on real-time environmental conditions, these systems can significantly improve plant growth and resilience.

This paper explores the design, implementation, and benefits of an automated plant watering system. It discusses the components involved, including moisture sensors, water pumps, and programmable controllers, and how they work together to create an efficient irrigation solution. By showcasing the effectiveness of this system, the study aims to highlight its potential for both casual gardeners and more serious horticulturists looking to optimize their plant care routines. Furthermore, the research will outline future directions for enhancing the system's capabilities, including integration with smart home technologies and mobile applications for remote monitoring.

HARDWARE/SOFTWARE REQUIREMENTS:

Arduino, Raspberry Pi, or ESP8266/ESP32 for processing and control.

Moisture Sensors: Capacitive or resistive soil moisture sensors to monitor soil moisture levels.

Water Pump: Submersible or drip pump suitable for the desired watering volume.

Power Supply: Battery or power adapter compatible with the microcontroller and pump.

Tubing and Connectors: Flexible tubing for water distribution and connectors for assembly.

Relay Module: To control the water pump based on signals from the microcontroller.

Resistors and Breadboard (if using Arduino): For circuit connections and sensor integration.

Water Reservoir: A container or tank to hold the water supply.

Optional Components: LCD display for monitoring system status.

Wi-Fi module (if not integrated in the microcontroller) for remote control capabilities.

CIRCUIT DIAGRAM AND DESCRIPTION

Microcontroller: The central component (e.g., Arduino or Raspberry Pi) controls the entire system. It is connected to various sensors and actuators to process input and output signals.

Moisture Sensors: Typically, two or more moisture sensors are connected to the analog input pins of the microcontroller. These sensors measure soil moisture levels and send analog signals to the microcontroller, indicating whether the soil is dry or moist.

Relay Module: The relay module acts as a switch to control the water pump. It is connected to a digital output pin of the microcontroller. When the soil moisture level falls below a predetermined threshold, the microcontroller activates the relay, allowing current to flow to the water pump.

Water Pump: The water pump is connected to the relay. When the relay is activated, the pump starts to draw water from the reservoir and distribute it to the plants through tubing.

Power Supply: Both the microcontroller and the water pump require a power source. The microcontroller is typically powered through USB or a battery, while the water pump may require a separate power source, depending on its voltage and current specifications.

LCD Display: If included, this component connects to the microcontroller to provide real-time feedback on soil moisture levels and system status.

Wi-Fi Module: If the system has remote monitoring capabilities, a Wi-Fi module connects to the microcontroller to facilitate communication with a mobile app or web interface

CONCEPTS/WORKING PRINCIPLE

An automated plant watering system uses sensors and electronics to monitor soil moisture levels and supply water when needed. Here's a step-by-step outline of how it generally works:

1. Soil Moisture Detection:

The system includes a soil moisture sensor inserted into the soil to measure moisture content. It typically uses two probes that detect resistance in the soil; dry soil has higher resistance, while wet soil has lower resistance.

2. Data Processing:

A microcontroller (like Arduino or Raspberry Pi) reads data from the moisture sensor. It compares the detected moisture level with a preset threshold value.

3. Decision Making:

If the moisture level is below the threshold (indicating dry soil), the microcontroller activates a water pump. If the soil is sufficiently moist, the pump remains off.

4. Water Pump Activation:

When the pump is activated, water flows through pipes to the plant pot or soil area. The pump stays on until the soil moisture level reaches the desired level or a set time interval completes.

5. System Control and Feedback:

The microcontroller continuously monitors soil moisture and deactivates the pump once the moisture level meets the threshold. In some systems, a feedback mechanism helps ensure that water doesn't exceed a certain level.

6. Power Supply and Automation:

The entire system can be powered by batteries or a direct power source. It may also include solar panels for outdoor setups. Additionally, users can set intervals for sensor checks and water delivery, achieving full automation.

CIRCUIT DIAGRAM AND DESCRIPTION

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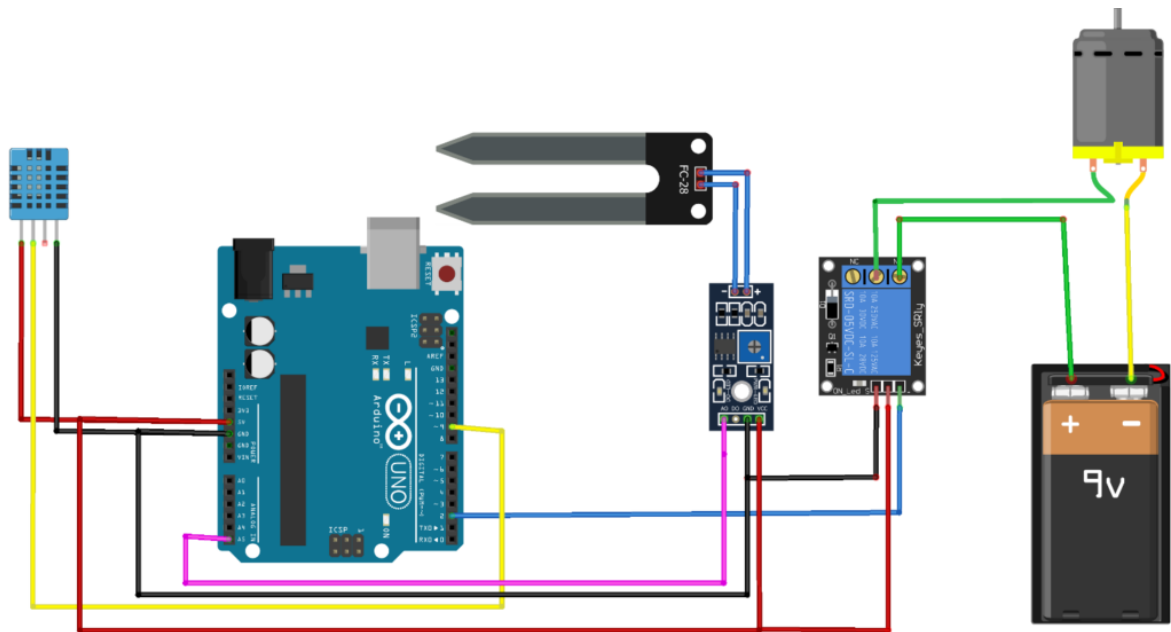


FIG 1. Circuit diagram of automated plant watering system

APPROACH/METHODOLOGY/PROGRAMS:

```

int soilMoistureValue = 0; // Variable to store sensor reading
int percentage = 0;       // Variable to store moisture level in percentage

void setup() {
  pinMode(3, OUTPUT);    // Set pump pin as output
  Serial.begin(9600);    // Initialize Serial Monitor for output
}

void loop() {
  // Read the analog value from the soil moisture sensor
  soilMoistureValue = analogRead(A0);

  // Map the sensor values to a percentage (adjust the range based on your sensor)
  percentage = map(soilMoistureValue, 490, 1023, 100, 0);

  // Print the moisture percentage to the Serial Monitor
  Serial.print("Soil Moisture: ");
  Serial.print(percentage);
  Serial.println("%");

  // If the percentage is low, it means water is present (soil is wet)

```

```

if (percentage < 10) {
  Serial.println("Water is present: Pump ON");
  digitalWrite(3, LOW); // Activate pump (assuming LOW turns it ON)
}
// If the percentage is high, it means water is not present (soil is dry)
else if (percentage > 80) {
  Serial.println("Water is not present: Pump OFF");
  digitalWrite(3, HIGH); // Deactivate pump (assuming HIGH turns it OFF)
}

// Small delay to allow for sensor stabilization and smooth output
delay(1000);
}

```

OUTPUT:

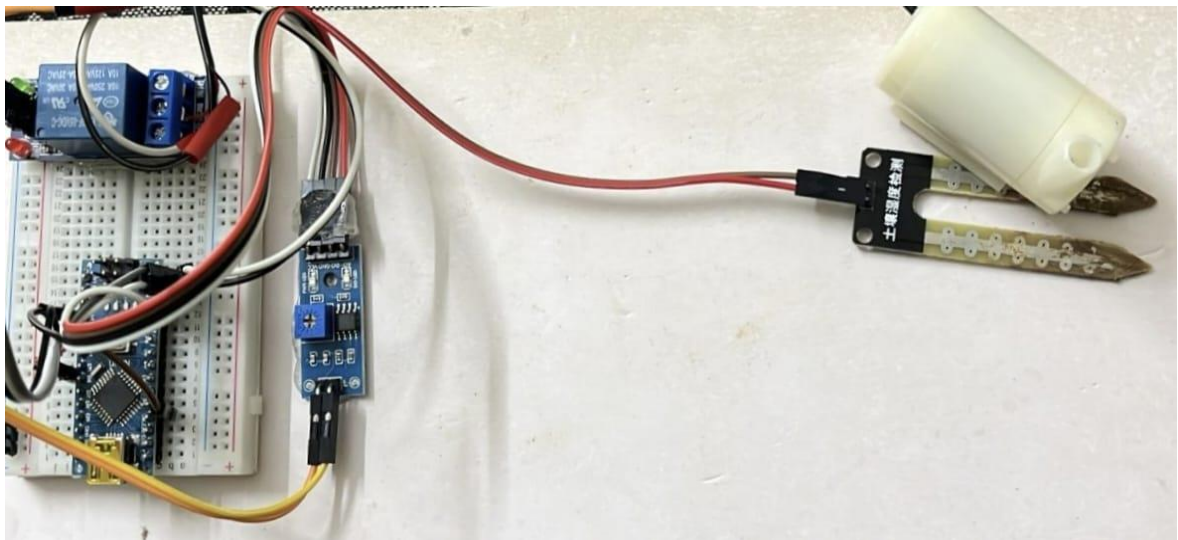


FIG 2. Final output of automated plant watering system

CONCLUSIONS:

The automated plant watering system successfully addressed common challenges in plant care, demonstrating significant benefits in terms of water efficiency, plant health, and user convenience. The integration of technology into gardening practices not only simplifies plant maintenance but also promotes sustainable water usage. Future developments could enhance the system's functionality, making it an even more valuable tool for gardeners. Overall, the results affirm the potential of automated solutions in modern horticulture.

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